# HTML MAANG Q&A

Top 15 Technical Interview Questions for Frontend Engineering

**Preparation Tip**

MAANG interviewers look beyond basic definitions. Always frame your answers in terms of *trade-offs, performance impacts,* and *accessibility (a11y)* to demonstrate senior-level understanding.

## SEMANTICS & STRUCTURE

### 1. Why is semantic HTML important?

Semantic HTML uses tags that convey the actual meaning of the content (e.g., `<article>`, `<nav>`) rather than just its presentation (`<div>`, `<span>`). Key benefits include:

- **Accessibility (a11y):** Screen readers rely on semantic tags to interpret page structure and provide navigation shortcuts.
- **SEO:** Search engine crawlers weight content inside semantic tags heavier, improving page ranking and indexing.
- **Maintainability:** It requires less custom CSS/JS to recreate native browser behaviors (like keyboard focus on a `<button>`).

### 2. Difference between <b>/<i> and /?

- `<b>` (bold) and `<i>` (italic) are **presentational** tags. They change visual appearance but add no structural meaning.
- `<strong>` (importance) and `<em>` (emphasis) are **semantic** tags. Screen readers change their tone/inflection when reading them.

### 3. Explain the difference between <section> and <article>.

- `<article>` represents a complete, self-contained piece of content that could be distributed independently (e.g., a blog post, a news story).
- `<section>` is a thematic grouping of content, usually requiring a heading (e.g., chapters in a book, tabs). You can have sections within an article, and vice versa.

## PERFORMANCE & CRITICAL RENDERING PATH

### 4. What is the difference between defer and async scripts?

Standard scripts block HTML parsing. Both defer and async download in the background without blocking, but differ in execution:

- **async:** Executes the exact moment it finishes downloading, pausing HTML parsing. Order is not guaranteed. Good for independent scripts (analytics).
- **defer:** Downloads in the background but waits to execute until HTML is fully parsed (before DOMContentLoaded). Executes in document order. Good for DOM-dependent scripts.

### 5. What is the Critical Rendering Path (CRP)?

The sequence of steps the browser takes to convert HTML, CSS, and JS into pixels:

1. Parse HTML to build the **DOM** (Document Object Model).
2. Parse CSS to build the **CSSOM** (CSS Object Model).
3. Combine DOM and CSSOM to create the **Render Tree**.
4. **Layout** (Reflow): Calculate exact sizes and positions.
5. **Paint**: Draw the pixels to the screen.

### 6. Optimize resource loading using <link rel="...">?

- `preload`: Download a critical resource immediately because it is needed for the current navigation.
- `prefetch`: Download a resource in the background because it *might* be needed for a future navigation.
- `preconnect`: Resolve DNS, TCP, and TLS handshakes early for a third-party domain.

## ACCESSIBILITY (A11Y)

### 7. What is ARIA and the First Rule of ARIA?

ARIA (Accessible Rich Internet Applications) defines ways to make web content more accessible to people with disabilities.

**First Rule:** No ARIA is better than bad ARIA. Always use native semantic HTML elements if they exist. Only use ARIA for custom UI components lacking native equivalents.

### 8. How do you ensure a custom component is keyboard accessible?

- Use `tabindex="0"` to make a non-focusable element focusable via the Tab key.
- Listen for keyboard events (`keydown`), specifically handling *Enter* and *Space* keys.
- Ensure distinct focus styling via CSS `:focus-visible`.

## SECURITY

### 9. What is Cross-Site Scripting (XSS) and HTML's role?

XSS occurs when an attacker injects malicious executable scripts into another user's browser.

**Prevention:** Never trust user input. Always escape/sanitize user-generated content. Avoid using `innerHTML` if you don't control the source; use `textContent` or `innerText` which treat content as raw text.

### 10. What is a Content Security Policy (CSP)?

An added layer of security that helps detect/mitigate attacks like XSS. Implemented via HTTP header or a `<meta>` tag, it explicitly whitelists the domains the browser is allowed to load resources from.

### 11. Why use rel="noopener noreferrer" on target="_blank"?

- **Security:** Prevents a malicious destination page from redirecting the original tab using `window.opener.location`.
- **Performance:** Prevents the new tab's JS from slowing down your original page thread.

## BROWSER STORAGE & APIS

### 12. Compare LocalStorage, SessionStorage, and Cookies.

- **Cookies:** Small (4KB). Sent automatically with every HTTP request. Good for auth tokens.
- **LocalStorage:** Larger (~5MB). Stored persistently on the client-side. Not sent to the server.
- **SessionStorage:** Same as LocalStorage, but cleared when the tab/window is closed.

### 13. What is the Shadow DOM?

An API that attaches an encapsulated DOM tree to an element. Styles and scripts inside do not leak out, and external CSS does not bleed in. Core pillar of Web Components.

### 14. How does native HTML5 form validation work?

HTML5 provides attributes like `required`, `pattern`, and `type="email"`. The browser natively checks constraints on submit. Hook into this via the Constraint Validation API (`checkValidity()`) and style with `:valid` / `:invalid`.

### 15. Difference between alt and title attributes on images?

`alt` provides screen reader descriptions and fallback text. `title` offers tooltip advisory info (not reliably accessible).